Artificial Intelligence: Explaining the Basics

If you are a student or professional interested in the latest trends in the computing world, you would have heard of terms like artificial intelligence, data science, machine learning, deep learning, etc. The first article in this series on artificial intelligence explains these terms, and sets the platform for a simple tutorial that will help beginners get started with AI.

Today it is absolutely necessary for any student or professional in the field of computer science to learn at least the basics of AI, data science, machine learning and deep learning. However, where does one begin to do so?

To answer this question, I have gone through a number of textbooks and tutorials that teach AI. Some start at a theoretical level (a lot of maths), some teach you AI in a language-agnostic way (they don't care whether you know C, C++, Java, Python, or some other programming language), and yet others assume you are an expert in linear algebra, probability, statistics, etc. In my opinion, all of them are useful to a great extent. But the important question remains — where should an absolute beginner interested in AI begin his or her journey?

Frankly, there are many fine ways to begin your AI journey. However, I have a few concerns regarding many of them. Too much maths is a distraction while too little is like a driver who doesn't know where his/her car's engine is placed. Starting with advanced concepts is really effective if the potential AI engineer or data scientist is good in linear algebra, probability, statistics, etc. Starting with the very basics and ending at the middle of nowhere is fine only if the potential AI learner wants to end the journey at that particular point. Considering all these facts, I believe an AI tutorial for beginners should start at the very basics and end with a real AI project (small, yet one that will outperform any conventional program capable of performing the same task).

This series on AI will start from the very basics and will reach up to an intermediate level. But in addition to discussing topics in AI, I also want to 'cut the clutter' (the name of a popular Indian news show) about the topics involved since there is a lot of confusion among people where terms like AI, machine learning, data science, etc, are concerned. AI based applications are often necessary due to the huge volume of data being produced every single day. A cursory search on the Internet will tell you that about 2.5 quintillion bytes of data are produced a day (quintillion is the enormous number 10^{18}). However, do remember that most of this data is absolutely irrelevant to us, including tons of YouTube videos with no merit, emails sent without a second thought, reports on trivial matters on newspapers, and so on and so forth. However, this vast ocean of data also contains invaluable knowledge